Skills Summary

Getting the Vector, the Scale Factor and the Centre Right

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding a translation vector

Rule: $\langle a,\ b\rangle = \langle x' - x,\quad y' - y\rangle$ — image *minus* pre-image, always in that order.

- Reversing the order gives the *inverse* translation, the one that sends the image back to the original.
- Check by applying it: $(x + a,\ y + b)$ must land on the image.

Example: A translation maps $A(-2,\ 3)$ to $A'(6,\ -1)$. Find the vector.
Step 1. The vector records the trip from pre-image to image, so each component is the image coordinate minus the pre-image coordinate — set that subtraction up before evaluating anything.
Step 2. Horizontal: $6 - (-2) = \mathbf{8}$. Vertical: $-1 - 3 = \mathbf{-4}$. The vector is eight right and four down.

Try it: A translation maps $B(4,\ -5)$ to $B'(-1,\ 2)$. Find the vector.

check: **−5** and **7**

Skill 2 — Finding a scale factor

Rule: $k = \dfrac{\text{distance from centre to image}}{\text{distance from centre to pre-image}}$ — a *ratio*, never a difference.

- $k > 1$ enlarges, $0 < k < 1$ shrinks, $k = 1$ changes nothing.
- Measure both distances from the *same* centre, or the ratio means nothing.

Example: A dilation centred at the origin maps $P(3,\ 4)$ to $P'(6,\ 8)$. Find k.
Step 1. A scale factor compares distances from the centre, so measure both distances first and divide — the coordinates themselves are not the comparison.
Step 2. $OP = \sqrt{9 + 16} = \mathbf{5}$ and $OP' = \sqrt{36 + 64} = \mathbf{10}$.
Step 3. $k = \dfrac{10}{5} = \mathbf{2}$.

Try it: A point lies 13 units from the centre and its image lies 39 units from the same centre. Find k.

check: **3**

(!) Watch out: Subtracting the two distances, or two matching coordinates, gives a number that changes if you rescale the whole picture. A scale factor must not.

Skill 3 — Dilating about a given centre

Rule: $A' = C + k\,(A - C)$, one coordinate at a time: $x' = u + k(x - u)$ and $y' = v + k(y - v)$, where $C = (u,\, v)$.

- Multiplying the coordinates by k is the shortcut for C at the *origin* only.
- The centre is the one point that does not move — so it lies on every line joining a point to its image.

Example: Dilate $A(3,\, 5)$ by $k = 3$ about $C(1,\, 2)$.
Step 1. The centre is not the origin, so the coordinates cannot simply be multiplied — step out from the centre, scale that step, and step back.
Step 2. $x' = 1 + 3(3 - 1) = 1 + 6 = \mathbf{7}$ and $y' = 2 + 3(5 - 2) = 2 + 9 = \mathbf{11}$.

Try it: Dilate $B(1,\, 6)$ by $k = 2$ about $C(-2,\, 4)$.

check: **4** and **8**